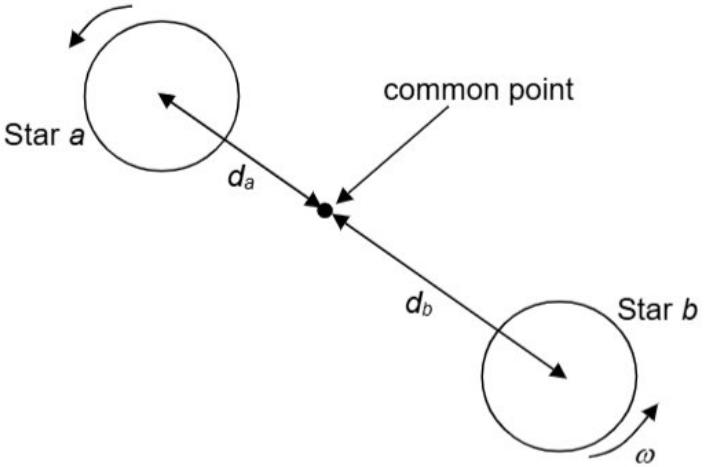
11 A binary star is a stellar system consisting of 2 stars of equal mass m orbiting around a common point, as shown below.



The distance between Star *a* and the common point is d_a while the distance between Star *b* and the common point is d_b.

Which of the following gives an expression for ω the angular velocity of Star *b*?

A $\sqrt{\dfrac{Gm}{(d_a+d_b)^3}}$ **B** $\dfrac{1}{d_a+d_b}\sqrt{\dfrac{Gm}{d_a}}$ **C** $\sqrt{\dfrac{Gm}{d_b^3}}$ **D** $\dfrac{1}{d_a+d_b}\sqrt{\dfrac{Gm}{d_b}}$